Supplementary Methods and Results

This supplement belongs to P11-MANUSCRIPT-CLOSEOUT-1, not original R3. All results are copied from exposed development evidence; independent validation was not executed.

S1. Full Configuration Grid

All 57 configurations are retained below. F = family, K/d = number of states/dimension, L = extra history lags, P = patient-weighted mean nominal coefficients INCLUDING shared offsets. Scores are development nats/unit-bin. Baseline names identify their distinct covariate sets. Full precision and training-objective seed policy remain in all57configurations.csv.

Family	Ridge	K/d	L	P total	Development score
stationary	0.0	-	-	23.389	-0.541291238
history	0.001	-	-	140.333	-0.541148902
history	0.01	-	-	140.333	-0.625239693
history	0.1	-	-	140.333	-0.863300484
history	1.0	-	-	140.333	-1.036731430
event	0.001	-	-	163.722	-0.545506185
event	0.01	-	-	163.722	-0.602585421
event	0.1	-	-	163.722	-0.813795828
event	1.0	-	-	163.722	-1.021970414
drift	0.001	-	-	46.778	-0.541583493
drift	0.01	-	-	46.778	-0.579602241
drift	0.1	-	-	46.778	-0.802580248
drift	1.0	-	-	46.778	-1.023682681
combined	0.001	-	-	304.056	-0.534921315
combined	0.01	-	-	304.056	-0.568088596
combined	0.1	-	-	304.056	-0.760691587
combined	1.0	-	-	304.056	-0.995955080
population	0.001	-	-	3841.370	-0.541962913
population	0.01	-	-	3841.370	-0.554542315
population	0.1	-	-	3841.370	-0.662697017
population	1.0	-	-	3841.370	-0.910859720
discrete	0.001	2	0	353.833	-0.518112435
discrete	0.01	2	0	353.833	-0.547175973
discrete	0.1	2	0	353.833	-0.565546349
discrete	1.0	2	0	353.833	-0.567812260
discrete	0.001	4	0	412.611	-0.515393923
discrete	0.01	4	0	412.611	-0.554497068

Family	Ridge	K/d	L	P total	Development score
discrete	0.1	4	0	412.611	-0.566760152
discrete	1.0	4	0	412.611	-0.567949724
continuous	0.001	1	0	357.833	-0.550125007
continuous	0.01	1	0	357.833	-0.551795924
continuous	0.1	1	0	357.833	-0.568088507
continuous	1.0	1	0	357.833	-0.568088596
continuous	0.001	2	0	413.611	-0.540875774
continuous	0.01	2	0	413.611	-0.546674737
continuous	0.1	2	0	413.611	-0.568087082
continuous	1.0	2	0	413.611	-0.568088596
continuous	0.001	4	0	531.167	-0.537378019
continuous	0.01	4	0	531.167	-0.543501399
continuous	0.1	4	0	531.167	-0.568087527
continuous	1.0	4	0	531.167	-0.568088596
hybrid	0.001	2	1	400.611	-0.516518920
hybrid	0.01	2	1	400.611	-0.546237251
hybrid	0.1	2	1	400.611	-0.565489020
hybrid	1.0	2	1	400.611	-0.567808203
hybrid	0.001	2	5	587.722	-0.515997654
hybrid	0.01	2	5	587.722	-0.545279771
hybrid	0.1	2	5	587.722	-0.565413767
hybrid	1.0	2	5	587.722	-0.567805368
hybrid	0.001	4	1	506.167	-0.514453621
hybrid	0.01	4	1	506.167	-0.553912039
hybrid	0.1	4	1	506.167	-0.566732507
hybrid	1.0	4	1	506.167	-0.567948134
hybrid	0.001	4	5	880.389	-0.514053175
hybrid	0.01	4	5	880.389	-0.552932337
hybrid	0.1	4	5	880.389	-0.566703689
hybrid	1.0	4	5	880.389	-0.567947120

S2. Complete Patient Scores

The four columns reproduce the original selected/reference means. Baseline is combined ridge0.001, not the stationary one-SEbaseline. No model selection is repeated.

Patient	Baseline	Discrete	Continuous	Hybrid
sub-01	-0.440993096	-0.459340501	-0.483732157	-0.457733319
sub-02	-0.577737752	-0.589624930	-0.601118745	-0.587867109
sub-03	-0.674959645	-0.680181759	-0.689338536	-0.679404843
sub-04	-0.663997952	-0.688733226	-0.714915792	-0.688245846
sub-05	-0.368052066	-0.362101626	-0.373153888	-0.361910558
sub-06	-0.464314056	-0.486346416	-0.522423518	-0.485862444
sub-07	-0.434149053	-0.446110056	-0.459398996	-0.446095589
sub-08	-0.526356200	-0.541865905	-0.587442574	-0.540726125
sub-09	-0.663732013	-0.670279340	-0.681273162	-0.668289428

S3. Full Patient Paired Effects

Patient	D-C	D-H	C-H	D-B	C-B	H-B
sub-01	+0.024392	-0.001607	-0.025999	-0.018347	-0.042739	-0.016740
sub-02	+0.011494	-0.001758	-0.013252	-0.011887	-0.023381	-0.010129
sub-03	+0.009157	-0.000777	-0.009934	-0.005222	-0.014379	-0.004445
sub-04	+0.026183	-0.000487	-0.026670	-0.024735	-0.050918	-0.024248
sub-05	+0.011052	-0.000191	-0.011243	+0.005950	-0.005102	+0.006142
sub-06	+0.036077	-0.000484	-0.036561	-0.022032	-0.058109	-0.021548
sub-07	+0.013289	-0.000014	-0.013303	-0.011961	-0.025250	-0.011947
sub-08	+0.045577	-0.001140	-0.046716	-0.015510	-0.061086	-0.014370
sub-09	+0.010994	-0.001990	-0.012984	-0.006547	-0.017541	-0.004557

S4. Saved Population Intervals and Tests

Contrast	Mean effect	95% exploratory CI	Raw sign-flip p	Holm-six p	Patients
discrete minus continuous	+0.020913	[+0.013508, +0.029372]	0.00390625	0.02343750	9
discrete minus hybrid	-0.000939	[-0.001374, -0.000497]	0.00390625	0.02343750	9
continuous minus hybrid	-0.021851	[-0.030288, -0.014437]	0.00390625	0.02343750	9
discrete minus baseline	-0.012255	[-0.017749, -0.006137]	0.01171875	0.02343750	9
continuous minus baseline	-0.033167	[-0.045727, -0.020913]	0.00390625	0.02343750	9
hybrid minus baseline	-0.011316	[-0.016862, -0.005293]	0.01562500	0.02343750	9

All six contrasts use n=9patients;10000paired patient bootstrap replicas and512exact sign flips. Holm-six was applied once to the saved contrast set. These intervals condition on the selected configurations and do not repeat selection; they are not selection-adjusted confirmatory inference.

S5. Conditional Patient Intervals

36descriptive intervals resample whole trials within each retained session and average sessions equally. They condition on the patient and are not independent-population CIs. No new resampling is performed.

Family	Patient	Mean	Saved 95% CI
baseline	sub-01	-0.440993096	[-0.476683, -0.405906]
baseline	sub-02	-0.577737752	[-0.596863, -0.557963]
baseline	sub-03	-0.674959645	[-0.691574, -0.659626]
baseline	sub-04	-0.663997952	[-0.677649, -0.651370]
baseline	sub-05	-0.368052066	[-0.436361, -0.313315]
baseline	sub-06	-0.464314056	[-0.484248, -0.445650]
baseline	sub-07	-0.434149053	[-0.449913, -0.418161]
baseline	sub-08	-0.526356200	[-0.568811, -0.487928]
baseline	sub-09	-0.663732013	[-0.679164, -0.647392]
continuous	sub-01	-0.483732157	[-0.512114, -0.456172]
continuous	sub-02	-0.601118745	[-0.620371, -0.581538]
continuous	sub-03	-0.689338536	[-0.703979, -0.676375]
continuous	sub-04	-0.714915792	[-0.724262, -0.705987]
continuous	sub-05	-0.373153888	[-0.433328, -0.323755]
continuous	sub-06	-0.522423518	[-0.536285, -0.509524]
continuous	sub-07	-0.459398996	[-0.472387, -0.446283]
continuous	sub-08	-0.587442574	[-0.618408, -0.560329]
continuous	sub-09	-0.681273162	[-0.695793, -0.665585]
discrete	sub-01	-0.459340501	[-0.489906, -0.429578]
discrete	sub-02	-0.589624930	[-0.610321, -0.568663]
discrete	sub-03	-0.680181759	[-0.695051, -0.666530]
discrete	sub-04	-0.688733226	[-0.698742, -0.679140]
discrete	sub-05	-0.362101626	[-0.427541, -0.309541]
discrete	sub-06	-0.486346416	[-0.501101, -0.472646]
discrete	sub-07	-0.446110056	[-0.460011, -0.432108]
discrete	sub-08	-0.541865905	[-0.573074, -0.514457]
discrete	sub-09	-0.670279340	[-0.685641, -0.654020]
hybrid	sub-01	-0.457733319	[-0.488066, -0.428214]
hybrid	sub-02	-0.587867109	[-0.608198, -0.567178]
hybrid	sub-03	-0.679404843	[-0.694309, -0.665742]
hybrid	sub-04	-0.688245846	[-0.698261, -0.678690]
hybrid	sub-05	-0.361910558	[-0.427441, -0.309321]
hybrid	sub-06	-0.485862444	[-0.500577, -0.472199]
hybrid	sub-07	-0.446095589	[-0.459994, -0.432101]
hybrid	sub-08	-0.540726125	[-0.571339, -0.513724]
hybrid	sub-09	-0.668289428	[-0.683343, -0.652305]

S6. Conditional Session Intervals

92descriptive whole-trial intervals below;23sessions per each of four selected/reference models. Trials, sessions and neurons are not independent patients.

Family	Session	Mean	Saved 95% CI
baseline	sub-01/ses-01	-0.541013497	[-0.583420, -0.492624]
baseline	sub-01/ses-02	-0.340972695	[-0.396956, -0.288446]
baseline	sub-02/ses-01	-0.592942966	[-0.633432, -0.554199]
baseline	sub-02/ses-02	-0.492263428	[-0.514302, -0.472599]
baseline	sub-02/ses-03	-0.547654632	[-0.562741, -0.536419]
baseline	sub-02/ses-04	-0.678089983	[-0.736117, -0.616202]
baseline	sub-03/ses-01	-0.612153341	[-0.627020, -0.595624]
baseline	sub-03/ses-02	-0.733023695	[-0.778830, -0.694984]
baseline	sub-03/ses-03	-0.679701898	[-0.697448, -0.662140]
baseline	sub-04/ses-01	-0.677185977	[-0.698957, -0.657411]
baseline	sub-04/ses-02	-0.650809926	[-0.667773, -0.635656]
baseline	sub-05/ses-01	-0.368052066	[-0.436361, -0.313315]
baseline	sub-06/ses-01	-0.484599992	[-0.512796, -0.455967]
baseline	sub-06/ses-02	-0.462582698	[-0.494020, -0.436226]
baseline	sub-06/ses-03	-0.445759477	[-0.489582, -0.408748]
baseline	sub-07/ses-02	-0.522817580	[-0.542060, -0.501638]
baseline	sub-07/ses-03	-0.345480527	[-0.369596, -0.320330]
baseline	sub-08/ses-02	-0.458177576	[-0.616394, -0.317641]
baseline	sub-08/ses-03	-0.531856766	[-0.564862, -0.497467]
baseline	sub-08/ses-04	-0.502423399	[-0.527969, -0.476558]
baseline	sub-08/ses-05	-0.612967059	[-0.662256, -0.563678]
baseline	sub-09/ses-01	-0.680480461	[-0.702302, -0.655734]
baseline	sub-09/ses-02	-0.646983566	[-0.667242, -0.625653]
continuous	sub-01/ses-01	-0.578898598	[-0.611526, -0.541512]
continuous	sub-01/ses-02	-0.388565717	[-0.433376, -0.346277]
continuous	sub-02/ses-01	-0.611755865	[-0.647986, -0.577410]
continuous	sub-02/ses-02	-0.529628792	[-0.546929, -0.514999]
continuous	sub-02/ses-03	-0.573200002	[-0.588584, -0.563312]
continuous	sub-02/ses-04	-0.689890319	[-0.751147, -0.624251]
continuous	sub-03/ses-01	-0.621186218	[-0.632888, -0.607756]
continuous	sub-03/ses-02	-0.748974527	[-0.789665, -0.717323]
continuous	sub-03/ses-03	-0.697854862	[-0.714014, -0.682445]
continuous	sub-04/ses-01	-0.724936266	[-0.738931, -0.711603]

Family	Session	Mean	Saved 95% CI
continuous	sub-04/ses-02	-0.704895318	[-0.717724, -0.693539]
continuous	sub-05/ses-01	-0.373153888	[-0.433328, -0.323755]
continuous	sub-06/ses-01	-0.535883909	[-0.555643, -0.516041]
continuous	sub-06/ses-02	-0.525411686	[-0.546610, -0.506791]
continuous	sub-06/ses-03	-0.505974958	[-0.535793, -0.479953]
continuous	sub-07/ses-02	-0.557571169	[-0.574323, -0.540823]
continuous	sub-07/ses-03	-0.361226824	[-0.380957, -0.341005]
continuous	sub-08/ses-02	-0.500998873	[-0.617986, -0.399601]
continuous	sub-08/ses-03	-0.580904004	[-0.602333, -0.557817]
continuous	sub-08/ses-04	-0.585944582	[-0.602735, -0.569014]
continuous	sub-08/ses-05	-0.681922838	[-0.713763, -0.650638]
continuous	sub-09/ses-01	-0.693230365	[-0.714540, -0.669619]
continuous	sub-09/ses-02	-0.669315958	[-0.688043, -0.649390]
discrete	sub-01/ses-01	-0.555756859	[-0.591329, -0.514791]
discrete	sub-01/ses-02	-0.362924144	[-0.411315, -0.317763]
discrete	sub-02/ses-01	-0.600462536	[-0.638088, -0.564662]
discrete	sub-02/ses-02	-0.503937003	[-0.522998, -0.487268]
discrete	sub-02/ses-03	-0.560153144	[-0.576618, -0.549365]
discrete	sub-02/ses-04	-0.693947038	[-0.760284, -0.623122]
discrete	sub-03/ses-01	-0.615714977	[-0.628512, -0.601165]
discrete	sub-03/ses-02	-0.736715690	[-0.777103, -0.703433]
discrete	sub-03/ses-03	-0.688114611	[-0.705067, -0.671732]
discrete	sub-04/ses-01	-0.699375097	[-0.715147, -0.684437]
discrete	sub-04/ses-02	-0.678091356	[-0.691291, -0.666387]
discrete	sub-05/ses-01	-0.362101626	[-0.427541, -0.309541]
discrete	sub-06/ses-01	-0.499967286	[-0.521142, -0.478647]
discrete	sub-06/ses-02	-0.489014248	[-0.510776, -0.469881]
discrete	sub-06/ses-03	-0.470057713	[-0.502578, -0.441645]
discrete	sub-07/ses-02	-0.540346851	[-0.557678, -0.522319]
discrete	sub-07/ses-03	-0.351873260	[-0.373134, -0.329858]
discrete	sub-08/ses-02	-0.441827745	[-0.557882, -0.341781]
discrete	sub-08/ses-03	-0.548614077	[-0.572235, -0.523410]
discrete	sub-08/ses-04	-0.539610343	[-0.558073, -0.520988]
discrete	sub-08/ses-05	-0.637411454	[-0.672027, -0.603171]
discrete	sub-09/ses-01	-0.684412478	[-0.706903, -0.659539]
discrete	sub-09/ses-02	-0.656146201	[-0.675635, -0.635311]
hybrid	sub-01/ses-01	-0.553268632	[-0.588683, -0.512798]

Family	Session	Mean	Saved 95% CI
hybrid	sub-01/ses-02	-0.362198005	[-0.410356, -0.317264]
hybrid	sub-02/ses-01	-0.599107057	[-0.636735, -0.563309]
hybrid	sub-02/ses-02	-0.503762973	[-0.522916, -0.487010]
hybrid	sub-02/ses-03	-0.558294882	[-0.573978, -0.547896]
hybrid	sub-02/ses-04	-0.690303526	[-0.755297, -0.621373]
hybrid	sub-03/ses-01	-0.615574033	[-0.628355, -0.601048]
hybrid	sub-03/ses-02	-0.735487616	[-0.775919, -0.702279]
hybrid	sub-03/ses-03	-0.687152879	[-0.703954, -0.670803]
hybrid	sub-04/ses-01	-0.699010213	[-0.714660, -0.684146]
hybrid	sub-04/ses-02	-0.677481479	[-0.690655, -0.665778]
hybrid	sub-05/ses-01	-0.361910558	[-0.427441, -0.309321]
hybrid	sub-06/ses-01	-0.499444488	[-0.520638, -0.478253]
hybrid	sub-06/ses-02	-0.488467315	[-0.510104, -0.469404]
hybrid	sub-06/ses-03	-0.469675527	[-0.502054, -0.441308]
hybrid	sub-07/ses-02	-0.540320781	[-0.557641, -0.522344]
hybrid	sub-07/ses-03	-0.351870397	[-0.373125, -0.329866]
hybrid	sub-08/ses-02	-0.439543137	[-0.553273, -0.341692]
hybrid	sub-08/ses-03	-0.547714650	[-0.571158, -0.522684]
hybrid	sub-08/ses-04	-0.539218376	[-0.557597, -0.520642]
hybrid	sub-08/ses-05	-0.636428337	[-0.670797, -0.602416]
hybrid	sub-09/ses-01	-0.682392476	[-0.704400, -0.657995]
hybrid	sub-09/ses-02	-0.654186381	[-0.673403, -0.633676]

S7. Complete Session Coverage

Session	Units	Train	Development
sub-01/ses-01	25	28	9
sub-01/ses-02	18	28	9
sub-02/ses-01	17	28	9
sub-02/ses-02	23	25	8
sub-02/ses-03	19	25	8
sub-02/ses-04	7	27	9
sub-03/ses-01	15	26	8
sub-03/ses-02	16	24	8
sub-03/ses-03	16	30	10
sub-04/ses-01	38	28	9
sub-04/ses-02	37	28	9
sub-05/ses-01	6	25	8
sub-06/ses-01	31	29	9
sub-06/ses-02	34	30	10
sub-06/ses-03	35	28	9
sub-07/ses-02	19	24	8
sub-07/ses-03	6	29	9
sub-08/ses-02	34	25	8
sub-08/ses-03	40	27	9
sub-08/ses-04	51	27	9
sub-08/ses-05	65	29	9
sub-09/ses-01	20	27	9
sub-09/ses-02	20	25	8

S8. Optimizer Diagnostics and Computational Cost

All 3887records remain in optimizer_diagnostics.csv, including all seeds. One-iteration records:117; largest recorded gradient infinity norm:0.0565088103506. First-unit design rank is not joint rank. Convergence does not mean small gradients or global optima.

Termination message	Recorded fits
CONVERGENCE: RELATIVE REDUCTION OF F <= FACTR*EPSMCH	3319
CONVERGENCE: NORM OF PROJECTED GRADIENT <= PGTOL	545
Exact constant-rate maximum likelihood	23

The recorded fit_seconds=805.5225947985891 is cumulative training timing in the original environment. It is not a new speed benchmark. Penalization, bounds[-8,8],500iterations,ftol/gtol1e-5 are mathematical parts of the compared pipelines. Nominal coefficient budgets are unequal; the frontier reports that inequality rather than asserting exact complexity matching.

S9. Numerical, Null and External Failures

OriginalM4/NUM1/NUM2/NUM3 failures are immutable. NUM2 failed a controlled proposal oracle before real data. NUM3 independent quadrature consistency is empirical; registered worst-replica-plus1.96SD-and-reference-sensitivity criterion failed at131072particles. None establishes that all numerical evaluators must fail. These are controlled algorithmic observations, not hippocampal experiments.

The null plan required200replicas each of3types. Eight full count-permutation cohort replicas and2896savednullfits (including partial work) were preserved. No partial-null pvalue or completed full-null claim appears. Circular/task-event suites and external final validation were not executed. DANDI5unknown-support sessions were not deleted to pass. Subsequent OSF/Figshare admission audits were inconclusive.

S10. Evidence Traceability and Scope

Each of the seven figure sidecars lists source paths andSHA256 plus render-code hash. build_inputs.json and claim_map.json bind the document to frozen sources. Saved-only reproduction regenerates tables/figures/document formatting; it neither refits models nor reproduces unexecuted controls/external evaluation. Author contributions/funding/competing-interest/submission declarations remain unprovided, explicitly disclosed in the main paper.

Supplementary References

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